Duke University
Math 106L: Laboratory Calculus and Functions II

Exam 1 Practice

THE FOLLOWING INSTRUCTIONS WILL APPEAR ON THE EXAM:

- Do not open this test booklet until you are directed to do so.
- You will have 105 minutes to complete the exam. If you finish early go back and check your work.
- This exam is closed book. However, you may use one side of a letter-sized piece of paper with your own *hand written* notes.
- You may not use a calculator or any other technology.
- Throughout the exam, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- Where exact answers are possible, decimal approximations will result in points deducted.

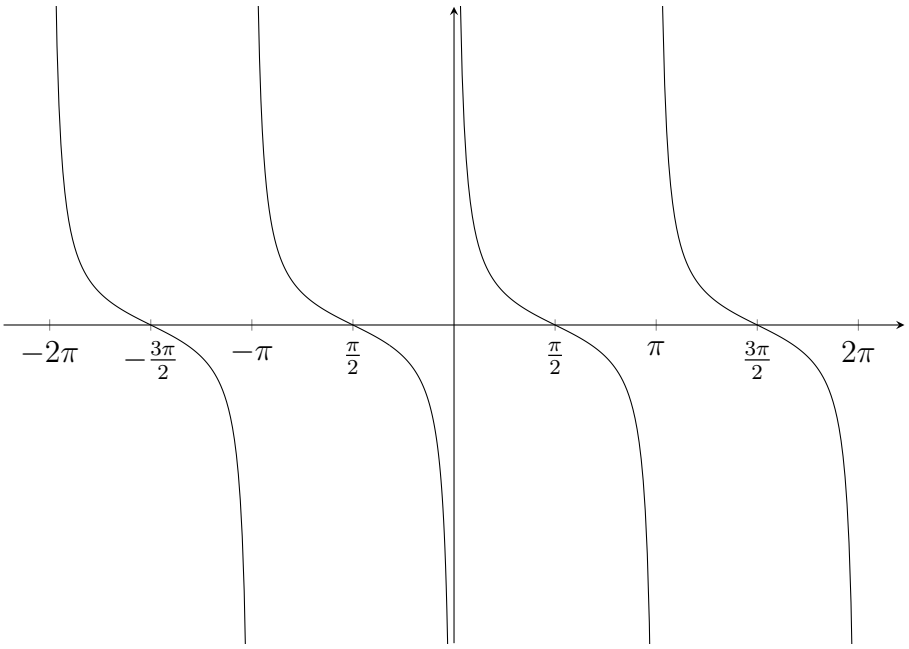
THE FOLLOWING TRIG IDENTITIES WILL BE GIVEN ON THE COVER OF THE EXAM:

Basic Trig Identities

$$\sin^2(x) + \cos^2(x) = 1$$

$$\begin{aligned}
\sin(A+B) &= \sin(A)\cos(B) + \cos(A)\sin(B) \\
\sin(A-B) &= \sin(A)\cos(B) - \cos(A)\sin(B) \\
\cos(A+B) &= \cos(A)\cos(B) - \sin(A)\sin(B) \\
\cos(A-B) &= \cos(A)\cos(B) + \sin(A)\sin(B)
\end{aligned}$$

1. Suppose that $f(x) = \cot(x)$. A graph of $f(x)$ with domain $[-2\pi, 2\pi]$ is shown below.



(a) Briefly explain why $f(x)$ is not invertible.

(b) What domain (containing positive numbers as close to zero as possible) should you restrict $f(x)$ to in order to make it invertible?

For the rest of this question, assume $f(x)$ is restricted as in part (b).

(c) What are the domain and range of $f^{-1}(x)$?

(d) Find $\lim_{x\to\infty} f^{-1}(x)$.

2. Suppose that $g(x) = 2\sin(x)\cos(x)$.

   (a) Show that $g(x) = \sin(2x)$.

   (b) Find $g'(x)$ in two ways:

      i. Using the product rule.

      ii. Using the identity you showed in part (a) above.

   (c) Use your answers to parts (b)(i) and (b)(ii) to find an identity involving $\cos(2x)$.

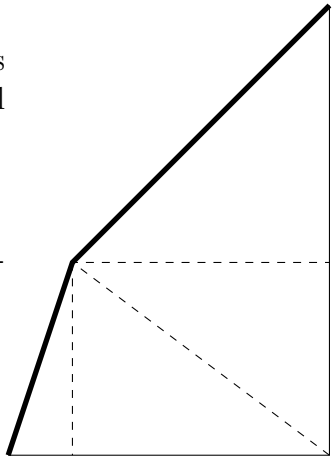
   $\cos(2x) =$ _____

   (d) Using your answer to part (c) or otherwise, show that $\cos(2x) = 1 - 2\sin^2(x)$.

3. You are standing some distance away from the launch site of a rocket. Both you and the launch site are on a flat plain.

   The rocket takes off at an angle of $75°$ from the horizontal. It climbs for a while, then lowers its angle of ascent to $55°$. At this point in time, you measure the straight-line distance between you and the rocket, and find that it is exactly $2,000$ meters, and that the angle of elevation from you to the rocket is $35°$.

   (a) On the picture to the right, the thick line represents the flight path of the rocket. Label the picture with all the information given in the paragraph above.



   (b) Compute the following two distances, leaving your answer in terms of trig functions:

      i. How far you are from the launch site;

      ii. The altitude of the rocket when the rocket is directly up over your head.

4. Solve the following equations in the given domains, or show that there are no solutions:

   (a) $\sec(x) = 0.5,\ 0 \leq x \leq 2\pi$;

   (b) $3\cos^2(x) + \cos(x) = 2,\ 0 \leq x \leq 4\pi$;

   (c) $\cos^2(x) + \sin^2(x) = 1.5,\ -2\pi \leq x \leq 2\pi$;

   (d) $2\sin^2(x) + \cos^2(x) = 1,\ -3\pi \leq x \leq 5\pi$.

5. A 12 foot long ladder is leaning against a house. Its base starts to slip away from the house. By the time the base is 8 feet away from the house, the base is moving at a rate of 4 feet per second. Find the rate at which θ, the angle between the ground and the ladder, is changing at this time.

   (Hint: Draw a picture first!)

6. The Curiosity Rover is running around Mars. Its masters want to measure the acceleration due to gravity there, so they make the Rover drop throw a rock upward off a cliff. The rock's initial velocity is 10 meters per second, and the cliff is 2000 meters high. Let the acceleration due to gravity on Mars be m meters per second per second. Denote the height of the rock above the ground at time t seconds as $h(t)$, measured in meters.

   (a) Briefly explain why the following initial value problem models this situation:

$$h''(t) = -m \text{ meters per second per second}$$
$$h'(0) = 10 \text{ meters per second}$$
$$h(0) = 2000 \text{ meters}$$

   (b) Solve the initial value problem above to find an expression for $h(t)$. Your answer will involve m.

   (c) If the rock takes 35.4 seconds to hit the ground, show that the acceleration due to gravity on Mars (m) is approximately 3.75 meters per second per second. (Hint: once you have an equation involving m and t, you do not need to solve it, just to check that the values given satisfy it. Note: this problem requires a calculator – no problem on the real exam will!)

7. For each of the following, state whether it is true or false. If it is false, provide a counterexample or a reason it is false.

   (a) The function $f(x) = \arccos(x)$ has one or more vertical asymptotes.

   (b) The function $g(x) = |\cos(x)|$ has period 2π.

   (c) The function $h(x) = \arcsin(x)$ is invertible.

   (d) If $f'(x) = \sin(x)$, then $f(x)$ must be $-\cos(x)$.

   (e) The function $g(x) = -2\sin(6(x-4)) + 7$ has amplitude 2 and period $\frac{\pi}{3}$.